CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 9 of 10 (Start time: 02:42:37 PM, 10 November 2025)

The total running time for Selection algorithm is _____ in n.

Select the correct option

○	Geometric
●	Linear
○	Exponential
○	Quadratic

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 7 of 10 (Start time: 02:42:08 PM, 10 November 2025)

In Selection problem, the Sieve technique works in _____

Select the correct option

◉	Phases
○	Non-recursive manner
○	One complete go
○	Constant time

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 8 of 10 (Start time: 02:42:23 PM, 10 November 2025)

A Principal operation for maintaining the heap property is called heapify, it is also called:

Select the correct option

◉	Sifting Down
○	Sifting Up
○	Sifting Right
○	Sifting Left

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 4 of 10 (Start time: 02:41:27 PM, 10 November 2025)

While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

Select the correct option

○	x < y
○	x > y
○	x = y
●	All of the above

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 5 of 10 (Start time: 02:41:41 PM, 10 November 2025)

While solving Selection problem, in Sieve technique we partition input data _____

Select the correct option

○	In decreasing order
●	According to Pivot
○	In increasing order
○	Randomly

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 6 of 10 (Start time: 02:41:54 PM, 10 November 2025)

In Heap Sort algorithm, the first step is to _____.

Select the correct option

◉	Call Build-Heap procedure
○	Sort the array in descending order
○	Find the number of input elements
○	Call Heapify procedure

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 3 of 10 (Start time: 02:41:09 PM, 10 November 2025)

If input "n" is odd, then median will be _____

Select the correct option

◉	(n+1)/2
○	(n-1)/2
○	n + 2
○	n/2

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 2 of 10 (Start time: 02:40:56 PM, 10 November 2025)

In Sieve technique, we solve the problem _____

Select the correct option

○	Using Brute force technique
◉	In recursive manner
○	Using Merge Sort algorithm
○	Non recursively

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 10 of 10 (Start time: 02:40:01 PM, 10 November 2025)

Selection sort takes theta _____ in the worst case.

Select the correct option

○	(n)
◉	(n^2)
○	nlog(n)
○	log(n)

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 1 of 10 (Start time: 02:40:43 PM, 10 November 2025)

While analyzing algorithms, _____and _____ are usually considered difficult to calculate.

Select the correct option

○	Row, Column
◉	Floor, ceiling
○	Finite, Infinite
○	Graph, Tree

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 9 of 10 (Start time: 02:39:48 PM, 10 November 2025)

While solving Selection problem, in Sieve technique we partition input data _____

Select the correct option

◉	According to Pivot
○	In increasing order
○	Randomly
○	In decreasing order

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 8 of 10 (Start time: 02:39:29 PM, 10 November 2025)

The number of nodes in a complete binary tree of height h is:

Select the correct option

○	((h+1) ^ 2) – 1
◉	2^(h+1) – 1
○	2 * (h+1) – 1
○	2 * (h+1)

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 5 of 10 (Start time: 02:38:45 PM, 10 November 2025)

Sorting can be in _____.

Select the correct option

○	Random order
○	Increasing order only
●	Both Increasing and Decreasing order
○	Decreasing order only

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 6 of 10 (Start time: 02:39:00 PM, 10 November 2025)

While solving Selection problem, in Sieve technique we choose pivot _____

Select the correct option

○	Minimum element
○	Average element
●	Randomly
○	Maximum element

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 7 of 10 (Start time: 02:39:13 PM, 10 November 2025)

In Heap Sort algorithm, the total running time for Heapify procedure is _____.

Select the correct option

○	O (1) i.e. Constant time
○	Big-oh(log n)
●	Theta (log n)
○	Omega (log n)

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 4 of 10 (Start time: 02:38:28 PM, 10 November 2025)

In Heap Sort algorithm, to remove the maximum element every time,_____.

Select the correct option

◉	We call heapify procedure
○	We call Build-Heap procedure
○	Heap Sort algorithm terminates without result
○	Nothing happens

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 2 of 10 (Start time: 02:37:58 PM, 10 November 2025)

The sieve technique is a special case, where the number of sub-problems is just _____

Select the correct option

○	4
○	3
◉	1
○	2

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 3 of 10 (Start time: 02:38:11 PM, 10 November 2025)

In max-heap, largest element is stored at root node. Where is the smallest element stored?

Select the correct option

○	Right Node
●	Leaf Node
○	Left Node
○	Middle Node

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 1 of 10 (Start time: 02:37:28 PM, 10 November 2025)

In Selection algorithm, we assume pivot selection takes theta _____ running time.

Select the correct option

◉	n
○	n^3
○	log(n)
○	n^2

CS502 - Fundamentals of Algorithms (Quiz No 1)

Question # 10 of 10 (Start time: 02:42:51 PM, 10 November 2025)

Sorting can be in _____.

Select the correct option

○	Increasing order only
○	Random order
○	Decreasing order only
◉	Both Increasing and Decreasing order